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Monocarboxylate transporter and lactate metabolism

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Abstract For many years, lactate was considered as a metabolic waste product, and a cause of fatigue during exercise. However, lactate is now known as a carbohydrate fuel source, shuttled between cells and tissues. The movement of lactate across the plasma membrane is facilitated by a family of monocarboxylate transporters (MCTs). Among 14 identified MCT isoforms, MCT1 and MCT4 are present in the plasma membranes of skeletal muscle and are suggested to be related to exercise performance. Reports have shown that exercise training increases both MCT1 and MCT4 in skeletal muscle. This review will discuss the role of lactate as a fuel and exercise training-induced changes in lactate metabolism. In addition, studies of Thoroughbred horses, as models of elite athletes, are introduced.

Keywords : lactate, monocarboxylate transporter (MCT), exercise training

Lactate as a metabolic fuel

Lactate accumulates in skeletal muscle during exercise as intensity increases. It is reported that lactate concentration in human muscle reaches over 30 mmol/l after 30 s of maximal exercise on an isokinetic cycle ergometer¹. The classical view was that an increase in lactate concentration is caused by an insufficient O₂ supply, and it induces muscle fatigue because of the intracellular acidosis. However, studies have shown that lactate is produced under fully aerobic conditions^{2,3}, and its relation to muscle fatigue is questioned. At present, it is considered that lactate itself may not have any negative effect on muscle contraction^{4,5}. Robergs et al. reviewed evidences that there is no biochemical support for lactate production causing acidosis and muscle fatigue⁶.

Lactate accumulation in muscle during intense exercise occurs mostly because the maximal rate of glycolysis is much greater than that of oxidation in the mitochondria; lactate production is essential for sustaining intense exercise that helps muscle cells produce ATP from glycolysis⁶. For many years, accumulated lactate in muscle was considered as an end product, and could only be converted back to glucose in the liver via the Cori cycle⁷. However, several studies suggested that lactate conversion to glucose via hepatic gluconeogenesis contributes less than 20% of lactate removal during exercise^{8,9}. Brooks and colleagues have shown that oxidation is the major route of lactate disposal (75-80%) during exercise and recovery by tracing metabolic fate of ¹³C-labeled lactate^{10,11}. Thus, we now understand that intramuscular lactate is an intermediate of carbohydrate metabolism because it is converted to pyruvate and subsequently oxidized in mitochondria

for ATP production. Miller et al. reported that lactate infusion at 4 mM of blood lactate concentration during moderate-intensity exercise increased lactate oxidation and spared blood glucose¹². Therefore, lactate can be a very useful carbohydrate source for skeletal muscle, even compared to glucose. It is important to know that skeletal muscle is not only the site of lactate production, but of lactate oxidation. In 1984, Brooks provided the framework of the lactate shuttle hypothesis (now called the cell-to-cell lactate shuttle) that holds that lactate is transferred between tissues and within a tissue¹³. Since its introduction, this hypothesis has been supported by a number of studies using various experimental methods. For example, van Hall et al. reported that during 40 min of exercise in cross-country skiers, the arms release lactate whereas the legs take up lactate¹⁴. In addition, it must be noted that the heart oxidizes lactate as a primary fuel¹⁵, since the cardiac muscle is one of the most oxidative muscles in the body¹⁶. Accordingly, we now understand that lactate is shuttled from its site of production (glycolytic fibers) to neighboring cells (oxidative fibers) and other organs such as the heart, liver, kidney and brain¹⁷.

Monocarboxylate transporter (MCT)

It was previously assumed that lactate produced in skeletal muscle was freely diffusible. However, several studies provided convincing evidence that lactate traverses the plasma membrane via a mechanism involving specific transport proteins. Roth and Brooks^{18,19} demonstrated that lactate transport is concentration-dependent and a saturable process with hydrogen ion dependence, and stereoselective for L-lactate. Since the first monocarboxylate transporter (MCT)1 was cloned in 1994 by Garcia et al²⁰, 14 isoforms of MCT have been identified. In skeletal

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muscle, MCT1 and MCT4 are believed to be key lactate transporters²¹). MCT1 is a high-affinity lactate transporter ($K_m = 3.5 - 8.3$ mM) predominantly present in oxidative fibers, whereas MCT4 is a low-affinity lactate transporter ($K_m = 25 - 31$ mM) in glycolytic fibers²²) (Fig. 1). Based on K_m values, MCT1 facilitates the uptake of lactate into muscle cells with low lactate levels, while MCT4 facilitates the extrusion of lactate out of muscle cells with a high concentration. These two isoforms of MCTs enable the lactate shuttle to function during exercise in muscles; lactate is produced in glycolytic muscles and released by MCT4, and then taken into oxidative muscles by MCT1 (Fig. 2). The model of the lactate shuttle is well established not only in skeletal muscle, but also in the brain. In the brain, lactate is generated in astrocytes and exported to the neurons via MCT for oxidation²³). The importance of MCT in the lactate shuttle in the brain has been recently reported²⁴).

Effects of exercise training on MCT and lactate metabolism

MCT expression is known to change in accordance with metabolic demands in muscle. Many studies have demonstrated increases in MCT protein content following various types of training. Low-intensity exercise training, such as voluntary wheel running²⁵) and chronic electrical stimulation^{26,27}), increases the MCT1 protein level, but not MCT4. High-intensity exercise training, such as sprint

interval training^{28,29}) and strength training³⁰), increases both MCT1 and MCT4 protein levels. While many studies have investigated the effects of training on MCT, only a few studies have reported changes in MCT by a single bout of exercise. Despite limited research, MCT expression seems to be rapidly altered in response to exercise. Acute exercise has been shown to immediately increase MCT1 and MCT4 protein in rats³¹) and humans³²). Thus, training-induced increases in MCTs may result from the cumulative effects of transient increases with a single bout of exercise. However, several studies reported that high-intensity exercise³³) and electrical stimulation³⁴) acutely decreased MCT1 and MCT4 protein levels in the plasma membrane. These inconsistent results may be due to differences in exercise intensity, exercise duration, type of subjects, and/or sample preparation methods. Studies of muscle inactivity models, such as denervation³⁵) or hindlimb suspension³⁶), have shown that lactate transport is decreased with a concomitant decrease in the protein levels of MCT1 and MCT4. Similarly, increased levels of MCT1 and MCT4 proteins, after 6 weeks of sprint-interval training, return to baseline levels after 6 weeks of detraining in humans²⁸). Thus, muscle contractile activity appears to be important for the regulation of MCT, and high intensity is needed for up-regulating MCT4. However, it must be acknowledged that at least two human studies reported that there were no significant changes in MCT1 and MCT4 protein levels even after high-intensity training^{37,38}). Therefore, exercise-induced changes in MCT

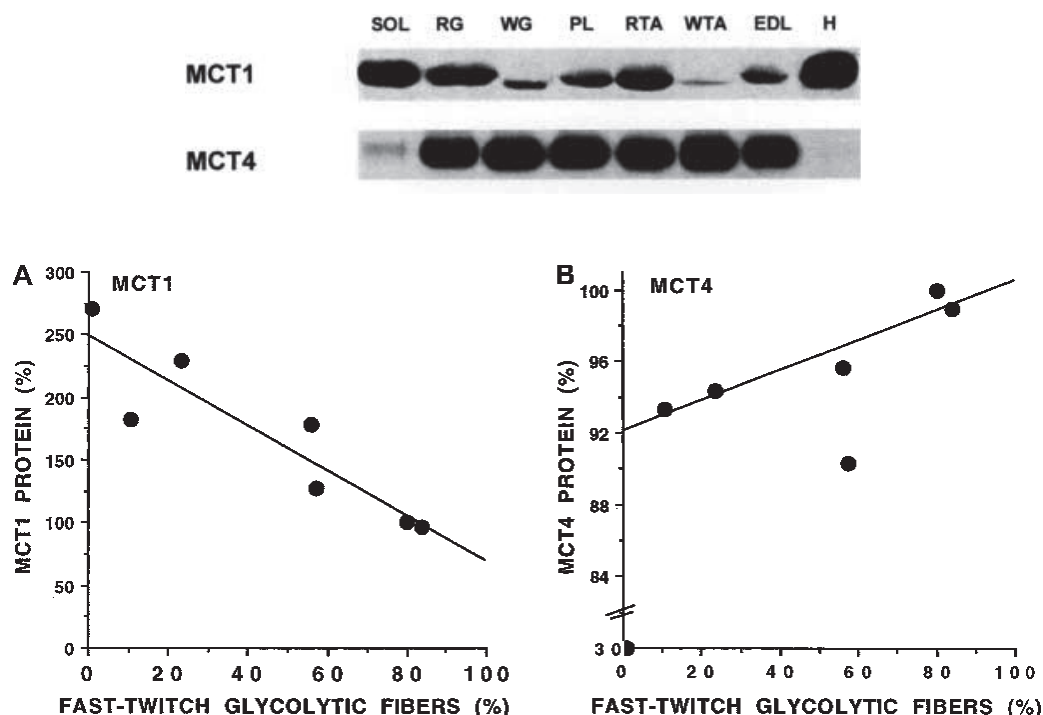


Fig. 1 Relationship between MCT1 (A) and MCT4 (B) proteins and muscle fiber composition (percentage of fast glycolytic fibers, %FG) in rat hindlimb muscles (SOL; soleus, PL; plantaris, EDL; extensor digitorum longus, RG and WG; red and white gastrocnemius, RTA and WTA; red and white tibialis anterior, H; heart). The MCT1 and MCT4 content of the white tibialis anterior was arbitrarily set to 100, and data from all other muscles are expressed relative to that of the white tibialis anterior. Note that among muscles there is a much wider MCT1 distribution than for MCT4. (Reference 21)

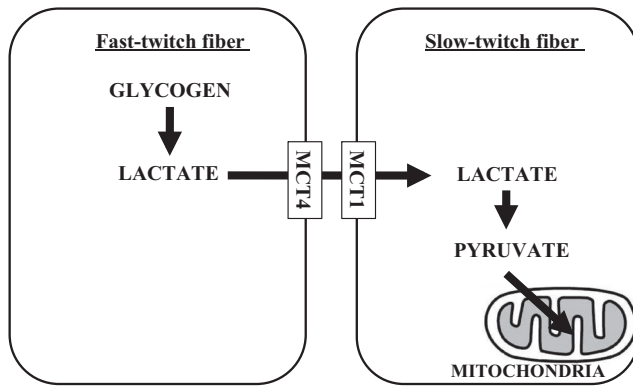


Fig. 2 Schematic overview of lactate shuttle. Lactate produced in glycolytic fibers is released to extra-cellular space and to blood by MCT4, and then taken up into oxidative fibers by MCT1. Thereafter, it is converted to pyruvate and oxidized in mitochondria for ATP synthesis.

may not only depend on training characteristics such as intensity and volume, but also training history and initial fitness of subjects; further study is needed to clarify whether well-trained athletes experience the same degree of increases in MCT after exercise as less-trained subjects.

Possible mechanisms for regulating MCT expression

Although many studies have investigated changes in MCT1 and MCT4 following various forms of training, the cellular and molecular mechanisms regulating MCT expression are still unknown. Accordingly, several hormones and drugs, and transcriptional factors, which regulate MCT expression, are reviewed here.

Triiodothyronine (T3) administration, which has profound effects on growth and development, was shown to increase the MCT4 protein level but not MCT1³⁹. Injection of testosterone, which is a principal male sex hormone and an anabolic steroid, increased MCT1 and MCT4 protein expression concomitantly with muscle hypertrophy⁴⁰. Additionally, clenbuterol (a beta 2-adrenergic agonist) administration increased the MCT4 protein level but decreased MCT1⁴¹. Taken together, MCT1 and MCT4 seem to respond differently to various hormones.

Recent studies suggest that MCT1 belongs to the family of metabolic genes whose expression is regulated by Peroxisome proliferator-activated receptor- γ coactivator-1 α (PGC-1 α), which is known as a master regulator of genes involved in oxidative capacity^{42,43}. Benton et al. reported a strong relationship between PGC-1 α and MCT1 proteins both in chronically stimulated and also PGC-1 α transfected muscles²⁶. These results are consistent with previous reports that MCT1 predominantly exists in oxidative fibers and is associated with oxidative capacity⁴⁴. Thus, an exercise training-induced increase in MCT1 is, at least in part, mediated by PGC-1 α .

Hypoxia is known to activate the expression of glycolytic enzymes via the transcriptional mechanism involving

hypoxia-inducible factor 1 α (HIF-1 α)⁴⁵. It is reported that HIF-1 α mediates MCT4 expression, but not MCT1⁴⁶. For example, chronic hypoxia up-regulated MCT4 in oxidative muscles⁴⁷. However, in contrast to laboratory animal studies, it is reported that there were no changes in MCT1 and MCT4 in Danish lowlander (human) muscle after 2- and 8-week stays at 4100 m in Bolivia⁴⁸. In addition, MCT4 protein does not always correlate with glycolytic enzyme activity⁴⁹. Therefore, it is possible that the training-induced increase in MCT4 expression is not only regulated by HIF-1 α , but also other factors. Together, MCT1 and MCT4 are regulated partially by PGC-1 α and HIF-1 α , respectively; but future studies are needed to understand the molecular mechanisms regulating MCT expression, using new technologies such as transgenic or knock-out animal models, of key molecules in major signal pathways.

Several reports suggest that MCT is regulated at various points during gene expression, since mRNA expression is not always correlated to protein levels. For example, it is reported that MCT1 protein expression increases by chronic electrical stimulation whereas mRNA does not²⁷. Similarly, both acute exercise and training increased MCT1 protein expression without increasing corresponding gene transcript levels⁵⁰. In contrast, T3 treatment up-regulated MCT1 mRNA, whereas protein expression remained unaltered³⁹. We recently found that the timing of functional overload-induced increases in MCT1 and MCT4 was different between mRNA and protein levels⁵¹. In addition, MCT1 protein to mRNA ratios were observed to differ greatly among different tissues⁵². These results suggest that the regulation of MCT expression seems to be primarily post-transcriptional.

Lactate metabolism in Thoroughbred horses as elite athletes

Thoroughbred horses are elite athletes, with a high capacity for high-intensity exercise. It is known that Thoroughbreds have high maximal oxygen uptake ($\dot{V}O_2$ max)⁵³ and a large amount of glycogen in the muscles⁵⁴. During maximal exercise in Thoroughbred horses, the muscle and plasma lactate concentrations increase to over 40 and 30 mmol/l, respectively⁵⁵. Since Thoroughbred horses have high aerobic and anaerobic capacities, they are considered suitable subjects for the study of lactate metabolism (Fig. 3). Studies in racehorses may provide new evidence that contributes towards improving exercise performance in human athletes.

It has been reported that MCT1 and MCT4 are present in equine muscles⁵⁶, and the level of MCT1, but not that of MCT4, increases with growth⁵⁷. Eighteen weeks of high-intensity training increased the muscle content of MCT1 and MCT4⁴⁹. In addition, it was demonstrated that high-intensity training is required to maintain the MCT4 protein level in well-trained horses. At present,



Fig. 3 Image of the incremental all-out test of the Thoroughbred horse in the Equine Research Institute, Japan Racing Association.

limited data in humans and other species are available regarding MCT protein levels and exercise performance. However, the authors previously reported the positive correlation between the MCT4 protein level and duration of the incremental all-out test in Thoroughbreds⁵⁸⁾, similar to MCT1. Furthermore, in another study, the authors found a positive relationship between the MCT4 protein level and increase in plasma lactate concentration during a period of incremental exercise at a moderate running speed until exhaustion (Fig. 4); this suggests that MCT4 facilitates lactate extrusion, particularly during high-intensity exercise. Thus, it can be expected that Thoroughbred horses sustain intense exercise, lasting a few minutes, with high glycolytic ATP production, since the muscle fibers remove lactate from contracting muscle at a very high rate via MCT4. Collectively, exercise performance, particularly high-intensity exercise, can be improved when the ability to remove lactate from glycolytic muscles is enhanced. Then, what is the fate of lactate? As mentioned throughout this review, lactate is a very useful carbohydrate source for oxidative muscles. It was found that the MCT1 protein level is highly correlated with citrate synthase activity throughout the training and detraining periods (Fig. 5). Therefore, increased lactate uptake by MCT1 and oxidative capacity can also be beneficial to exercise performance. Taken together, Thoroughbred horses are able to enhance both the removal of lactate from glycolytic fibers and utilization of lactate in oxidative fibers via MCT1 and MCT4.

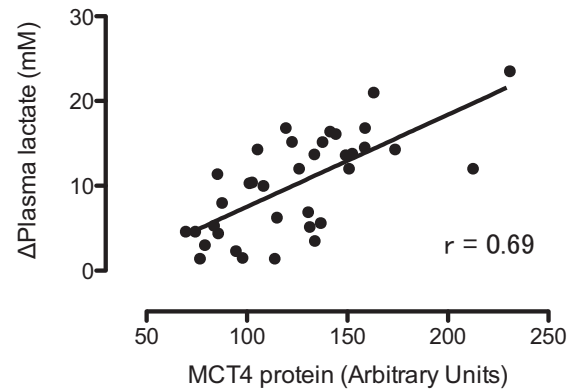


Fig. 4 Correlations between MCT4 protein content in equine gluteus medius muscle and the increase of plasma lactate concentration from the levels at a running speed of 10 m/s to those at exhaustion during the incremental all-out test. Data for all horses are included at baseline (Pre), after 18 weeks of high-intensity training, and after 6 weeks of detraining. (Reference 49)

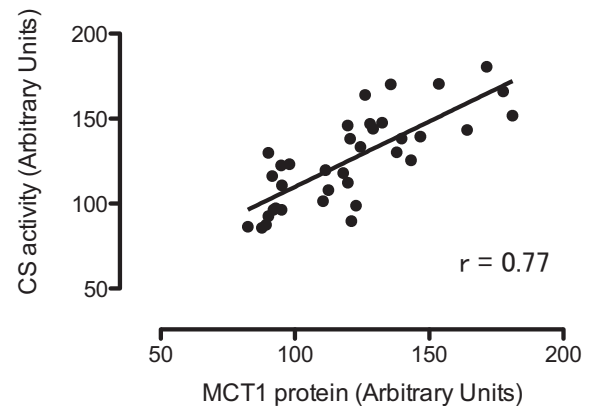


Fig. 5 Correlations between MCT1 protein content and citrate synthase activity in equine gluteus medius muscle. Data for all horses are included at baseline (Pre), after 18 weeks of high-intensity training, and after 6 weeks of detraining. (Reference 49)

References

- 1) Kowalchuk JM, Heigenhauser GJ, Lindinger MI, Sutton JR, Jones NL. 1988. Factors influencing hydrogen ion concentration in muscle after intense exercise. *J Appl Physiol* 65: 2080-2089.
- 2) Connett RJ, Gayeski TE, Honig CR. 1984. Lactate accumulation in fully aerobic, working, dog gracilis muscle. *Am J Physiol* 246: H120-H128.
- 3) Richardson RS, Noyszewski EA, Leigh JS, and Wagner PD. 1998. Lactate efflux from exercising human skeletal muscle: role of intracellular PO₂. *J Appl Physiol* 85: 627-634.
- 4) Allen D, Westerblad H. 2004. Lactic acid—the latest performance-enhancing drug. *Science* 305: 1112-1113.
- 5) Westerblad H, Bruton JD, Lannergren J. 1997. The effect of intracellular pH on contractile function of intact, single fibres of mouse muscle declines with increasing temperature. *J Physiol* 500 (Pt 1): 193-204.
- 6) Robergs RA, Ghiasvand F, Parker D. 2004. Biochemistry of exercise-induced metabolic acidosis. *Am J Physiol Regul In-*

- tegr Comp Physiol* 287: R502-R516.
- 7) Cori CF, Cori GT. 1929. Glycogen formation in the liver from d- and l-lactic acid. *J Biol Chem* 81: 389-403.
 - 8) Astrand PO, Hultman E, Juhlin-Dannfelt A, Reynolds G. 1986. Disposal of lactate during and after strenuous exercise in humans. *J Appl Physiol* 61: 338-343.
 - 9) Hermansen L, Vaage O. 1977. Lactate disappearance and glycogen synthesis in human muscle after maximal exercise. *Am J Physiol* 233: E422-E429.
 - 10) Brooks GA. 1986. The lactate shuttle during exercise and recovery. *Med Sci Sports Exerc* 18: 360-368.
 - 11) Mazzeo RS, Brooks GA, Schoeller DA, Budinger TF. 1986. Disposal of blood [1-13C]lactate in humans during rest and exercise. *J Appl Physiol* 60: 232-241.
 - 12) Miller B, Fattor J, Jacobs K, Horning M, Navazio F, Lindinger M, Brooks G. 2002. Lactate and glucose interactions during rest and exercise in men: effect of exogenous lactate infusion. *J Physiol* 544: 963-975.
 - 13) Brooks G. 2000. Intra- and extra-cellular lactate shuttles. *Med Sci Sports Exerc* 32: 790-799.
 - 14) Van Hall G, Jensen-Urstad M, Rosdahl H, Holmberg HC, Saltin B, Calbet JA. 2003. Leg and arm lactate and substrate kinetics during exercise. *Am J Physiol Endocrinol Metab* 284: E193-E205.
 - 15) Stanley WC. 1991. Myocardial lactate metabolism during exercise. *Med Sci Sports Exerc* 23: 920-924.
 - 16) Jansson E, Sylven C. 1986. Activities of key enzymes in the energy metabolism of human myocardial and skeletal muscle. *Clin Physiol* 6: 465-471.
 - 17) Gladden LB. 2008. A lactatic perspective on metabolism. *Med Sci Sports Exerc* 40: 477-485.
 - 18) Roth DA, Brooks GA. 1990. Lactate and pyruvate transport is dominated by a pH gradient-sensitive carrier in rat skeletal muscle sarcolemmal vesicles. *Arch Biochem Biophys* 279: 386-394.
 - 19) Roth DA, Brooks GA. 1990. Lactate transport is mediated by a membrane-bound carrier in rat skeletal muscle sarcolemmal vesicles. *Arch Biochem Biophys* 279: 377-385.
 - 20) Garcia CK, Goldstein JL, Pathak RK, Anderson RG, Brown MS. 1994. Molecular characterization of a membrane transporter for lactate, pyruvate, and other monocarboxylates: implications for the Cori cycle. *Cell* 76: 865-873.
 - 21) Bonen A. 2001. The expression of lactate transporters (MCT1 and MCT4) in heart and muscle. *Eur J Appl Physiol* 86: 6-11.
 - 22) Benton C, Campbell S, Tonouchi M, Hatta H, Bonen A. 2004. Monocarboxylate transporters in subsarcolemmal and intermyofibrillar mitochondria. *Biochem Biophys Res Commun* 323: 249-253.
 - 23) Pierre K, Pellerin L. 2005. Monocarboxylate transporters in the central nervous system: distribution, regulation and function. *J Neurochem* 94: 1-14.
 - 24) Suzuki A, Stern SA, Bozdagi O, Huntley GW, Walker RH, Magistretti PJ, Alberini CM. 2011. Astrocyte-neuron lactate transport is required for long-term memory formation. *Cell* 144: 810-823.
 - 25) Yoshida Y, Hatta H, Kato M, Enoki T, Kato H, Bonen A. 2004. Relationship between skeletal muscle MCT1 and accumulated exercise during voluntary wheel running. *J Appl Physiol* 97: 527-534.
 - 26) Benton CR, Yoshida Y, Lally J, Han XX, Hatta H, Bonen A. 2008. PGC-1 α increases skeletal muscle lactate uptake by increasing the expression of MCT1 but not MCT2 or MCT4. *Physiol Genomics* 35: 45-54.
 - 27) Bonen A, Tonouchi M, Miskovic D, Heddle C, Heikkilä JJ, Halestrap AP. 2000. Isoform-specific regulation of the lactate transporters MCT1 and MCT4 by contractile activity. *Am J Physiol Endocrinol Metab* 279: E1131-E1138.
 - 28) Burgomaster KA, Cermak NM, Phillips SM, Benton CR, Bonen A, Gibala MJ. 2007. Divergent response of metabolite transport proteins in human skeletal muscle after sprint interval training and detraining. *Am J Physiol Regul Integr Comp Physiol* 292: R1970-R1976.
 - 29) Perry CG, Heigenhauser GJ, Bonen A, Spriet LL. 2008. High-intensity aerobic interval training increases fat and carbohydrate metabolic capacities in human skeletal muscle. *Appl Physiol Nutr Metab* 33: 1112-1123.
 - 30) Juel C, Holten MK, Dela F. 2004. Effects of strength training on muscle lactate release and MCT1 and MCT4 content in healthy and type 2 diabetic humans. *J Physiol* 556: 297-304.
 - 31) Coles L, Litt J, Hatta H, Bonen A. 2004. Exercise rapidly increases expression of the monocarboxylate transporters MCT1 and MCT4 in rat muscle. *J Physiol* 561: 253-261.
 - 32) Green H, Halestrap A, Mockett C, O'Toole D, Grant S, Ouyang J. 2002. Increases in muscle MCT are associated with reductions in muscle lactate after a single exercise session in humans. *Am J Physiol Endocrinol Metab* 282: E154-E160.
 - 33) Bishop D, Edge J, Thomas C, Mercier J. 2007. High-intensity exercise acutely decreases the membrane content of MCT1 and MCT4 and buffer capacity in human skeletal muscle. *J Appl Physiol* 102: 616-621.
 - 34) Tonouchi M, Hatta H, Bonen A. 2002. Muscle contraction increases lactate transport while reducing sarcolemmal MCT4, but not MCT1. *Am J Physiol Endocrinol Metab* 282: E1062-E1069.
 - 35) McCullagh KJ, Bonen A. 1995. Reduced lactate transport in denervated rat skeletal muscle. *Am J Physiol* 268: R884-R888.
 - 36) Dubouchaud H, Granier P, Mercier J, Le Peuch C, Prefaut C. 1996. Lactate uptake by skeletal muscle sarcolemmal vesicles decreases after 4 wk of hindlimb unweighting in rats. *J Appl Physiol* 80: 416-421.
 - 37) Bishop D, Edge J, Thomas C, Mercier J. 2008. Effects of high-intensity training on muscle lactate transporters and postexercise recovery of muscle lactate and hydrogen ions in women. *Am J Physiol Regul Integr Comp Physiol* 295: R1991-R1998.
 - 38) Everts F, Medbo JJ, Bonen A. 2001. Effect of training intensity on muscle lactate transporters and lactate threshold of cross-country skiers. *Acta Physiol Scand* 173: 195-205.
 - 39) Wang Y, Tonouchi M, Miskovic D, Hatta H, Bonen A. 2003. T3 increases lactate transport and the expression of MCT4, but not MCT1, in rat skeletal muscle. *Am J Physiol Endocrinol Metab* 285: E622-E628.
 - 40) Enoki T, Yoshida Y, Lally J, Hatta H, Bonen A. 2006. Testosterone increases lactate transport, monocarboxylate transporter (MCT) 1 and MCT4 in rat skeletal muscle. *J Physiol* 577: 433-443.
 - 41) Hoshino D, Yoshida Y, Holloway GP, Lally J, Hatta H, Bonen A. 2012. Clenbuterol, a β_2 -adrenergic agonist, reciprocally alters PGC-1 α and RIP140 and reduces fatty acid and pyruvate oxidation in rat skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* 302: R373-R384.

- 42) Hashimoto T, Hussien R, Oommen S, Gohil K, Brooks GA. 2007. Lactate sensitive transcription factor network in L6 cells: activation of MCT1 and mitochondrial biogenesis. *FASEB J* 21: 2602-2612.
- 43) Lira VA, Benton CR, Yan Z, Bonen A. 2010. PGC-1 α regulation by exercise training and its influences on muscle function and insulin sensitivity. *Am J Physiol Endocrinol Metab* 299: E145-E161.
- 44) Hashimoto T, Masuda S, Taguchi S, Brooks G. 2005. Immunohistochemical analysis of MCT1, MCT2 and MCT4 expression in rat plantaris muscle. *J Physiol* 567: 121-129.
- 45) Mason S, Johnson RS. 2007. The role of HIF-1 in hypoxic response in the skeletal muscle. *Adv Exp Med Biol* 618: 229-244.
- 46) Ullah MS, Davies AJ, Halestrap AP. 2006. The plasma membrane lactate transporter MCT4, but not MCT1, is up-regulated by hypoxia through a HIF-1 α -dependent mechanism. *J Biol Chem* 281: 9030-9037.
- 47) Py G, Eydoux N, Lambert K, Chapot R, Koulmann N, Sanchez H, Bahi L, Peinnequin A, Mercier J, Bigard AX. 2005. Role of hypoxia-induced anorexia and right ventricular hypertrophy on lactate transport and MCT expression in rat muscle. *Metabolism* 54: 634-644.
- 48) Juel C, Lundby C, Sander M, Calbet JA, Hall G. 2003. Human skeletal muscle and erythrocyte proteins involved in acid-base homeostasis: adaptations to chronic hypoxia. *J Physiol* 548: 639-648.
- 49) Kitaoka Y, Masuda H, Mukai K, Hiraga A, Takemasa T, Hatta H. 2011. Effect of training and detraining on monocarboxylate transporter (MCT) 1 and MCT4 in Thoroughbred horses. *Exp Physiol* 96: 348-355.
- 50) Bickham DC, Bentley DJ, Le Rossignol PF, Cameron-Smith D. 2006. The effects of short-term sprint training on MCT expression in moderately endurance-trained runners. *Eur J Appl Physiol* 96: 636-643.
- 51) Kitaoka Y, Machida M, Takemasa T, Hatta H. 2011. Expression of monocarboxylate transporter (MCT) 1 and MCT4 in overloaded mice plantaris muscle. *J Physiol Sci* 61: 467-472.
- 52) Jackson VN, Price NT, Carpenter L, Halestrap AP. 1997. Cloning of the monocarboxylate transporter isoform MCT2 from rat testis provides evidence that expression in tissues is species-specific and may involve post-transcriptional regulation. *Biochem J* 324 (Pt 2): 447-453.
- 53) Jones JH, Longworth KE, Lindholm A, Conley KE, Karas RH, Kayar SR, Taylor CR. 1989. Oxygen transport during exercise in large mammals. I. Adaptive variation in oxygen demand. *J Appl Physiol* 67: 862-870.
- 54) Lindholm A, Bjerneld H, Saltin B. 1974. Glycogen depletion pattern in muscle fibres of trotting horses. *Acta Physiol Scand* 90: 475-484.
- 55) Harris P, Snow DH. 1988. The effects of high intensity exercise on the plasma concentration of lactate, potassium and other electrolytes. *Equine Vet J* 20: 109-113.
- 56) Koho NM, Hyypä S, Poso AR. 2006. Monocarboxylate transporters (MCT) as lactate carriers in equine muscle and red blood cells. *Equine Vet J Suppl* 36: 354-358.
- 57) Kitaoka Y, Hoshino D, Mukai K, Hiraga A, Takemasa T, Hatta H. 2011. Effect of growth on monocarboxylate transporters and indicators of energy metabolism in the gluteus medius muscle of Thoroughbreds. *Am J Vet Res* 72: 1107-1111.
- 58) Kitaoka Y, Wakasugi Y, Hoshino D, Mukai K, Hiraga A, Hatta H. 2010. Effects of high intensity training on monocarboxylate transporters in Thoroughbred horses. *Comp Exer Physiol* 6: 171-175.